Logarithmic Functions & Properties

Section 1: Definition of Logarithms and Inverses

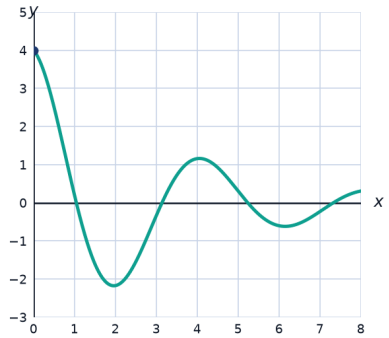
1 Evaluate $\log_4\left(\frac{1}{64}\right)$.

a. -3

b. 3

c. $-\frac{1}{3}$

d. 16



2 What is the domain of $f(x) = \ln(5 - 2x)$?

a. $(-\infty, 2.5)$

b. $(2.5, \infty)$

c. $(-\infty, 5)$

d. $(0, \infty)$

3 Find the inverse function of $g(x) = 3 + 2\ln(x - 1)$.

a. $g^{-1}(x) = e^{(x-3)/2} + 1$

b. $g^{-1}(x) = e^{(x-3)/2} - 1$

c. $g^{-1}(x) = 2e^{x-3} + 1$

d. $g^{-1}(x) = e^{2(x-3)} + 1$

Section 2: Logarithmic Properties & Condensing Expressions

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4 Express $2\log x - 3\log y + \frac{1}{2}\log z$ as a single logarithm.

a. $\log\left(\frac{x^2\sqrt{z}}{y^3}\right)$

b. $\log\left(\frac{2x\sqrt{z}}{3y}\right)$

c. $\log\left(x^2 - y^3 + \sqrt{z}\right)$

d. $\log\left(\frac{x^2}{y^3\sqrt{z}}\right)$

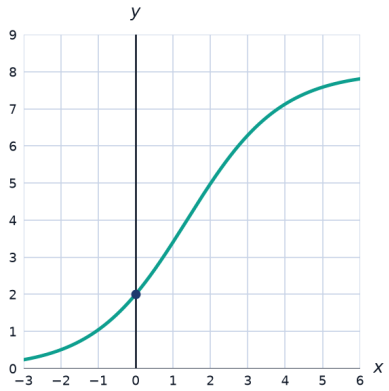
5 If $\log_b 2 = 0.35$ and $\log_b 3 = 0.55$, what is the value of $\log_b 18$?

a. 1.45

b. 0.90

c. 1.25

d. 1.10



6 Using the change of base formula, express $\log_5 12$ in terms of natural logarithms.

a. $\frac{\ln 12}{\ln 5}$

b. $\frac{\ln 5}{\ln 12}$

c. $\ln 12 - \ln 5$

d. $\ln\left(\frac{12}{5}\right)$

Section 3: Solving Exponential and Logarithmic Equations

7 Solve for x: $3^{2x-1} = 27$.

a. 2

b. 1

c. 3

d. $\frac{3}{2}$

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8 Solve for x: $\log_2(x) + \log_2(x-2) = 3$.

a. 4

b. 4 and -2

c. 2

d. 8

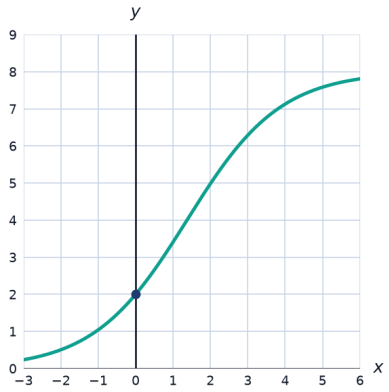
9 Solve for t: $50e^{0.04t} = 200$.

a. $\frac{\ln 4}{0.04} \approx 34.66$

b. $\frac{\ln 150}{0.04}$

c. $\frac{4}{0.04} = 100$

d. $\ln(4 \times 0.04)$



Section 4: Logarithmic Scales and Applications

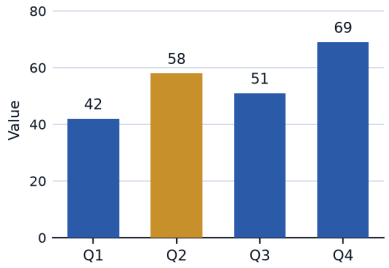
10 The pH of a chemical solution is given by $\text{pH} = -\log[\text{H}^+]$. If the hydrogen ion concentration is 1.0×10^{-5} M, what is the pH?

a. 5.0

b. -5.0

c. 1.0

d. 9.0



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11 An earthquake measures 7.2 on the Richter scale, while a second measures 5.2. How many times more intense is the first earthquake?

a. 100

b. 2

c. 20

d. 1000

12 What is the vertical asymptote of the graph of $f(x) = \log_3(x + 4) - 2$?

a. $x = -4$

b. $x = 4$

c. $y = -2$

d. $x = 2$

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